



Functions in C++

A function is a block of code that performs a specific task and can be reused anywhere in the program.

(ऐसा task/work जो बार-बार repeat हो रहा हैं तो उस task को function बनाकर use करते हैं)

User-defined Function

- A User defined function in C++ is a function that is define by the programmer to perform a specific task and can be called whenever needed in the program.

- **Syntax of User define function**

```
void hira()
```

```
{  
    Cout << "hello tourist" ;  
    Cout << "Welcome in Nalanda";  
}
```

// code to be executed

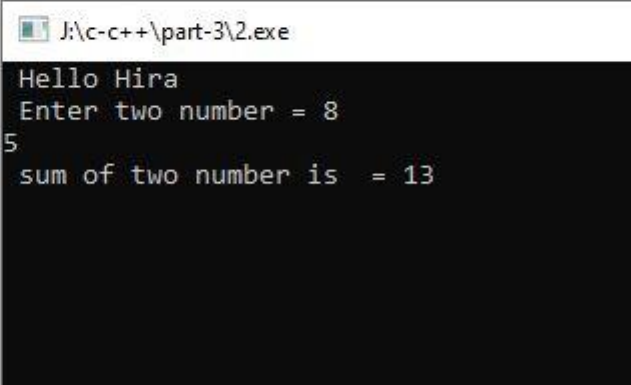
Here –

- ✓ hira() is the name of the function (User define).
- ✓ void means that the function does not have a return value.

Q1. WAP to add two number using function.

```
2.cpp X
#include<iostream>
#include<conio.h>
using namespace std;
void hira()
{
    int a,b;
    cout << " Enter two number = " ;
    cin >>a>>b;
    cout << " sum of two number is  = " <<a+b;
}

int main()
{
    cout << " Hello Hira" << endl;
    hira();
    getch();
}
```



❖ Function Declaration, Definition and call

- A **function declaration** tells the compiler about the function name, return type, and parameters before its actual definition.
It is also called the **function prototype**.
- A **function definition** contains the actual code that specifies what the function does.
- To **call a function**, write the function's name followed by two parentheses () and a semicolon ;

```

#include<iostream>
#include<conio.h>
using namespace std;
void hira(); → function declaration
int main()
{
    cout << " Hello Hira" << endl;
    hira(); → call the function
    getch();
}
void hira() → function definition
{
    int a,b;
    cout << " Enter two number = " ;
    cin >>a>>b;
    cout << " sum of two number is = " <<a+b;
}

```

Feature	Function Declaration (घोषणा)	Function Definition (परिभाषा)
Purpose	Compiler को advance में बताना	Function का असली काम करना
Ends with semicolon	✓ हाँ	✗ नहीं
Body (Code Block)	✗ नहीं होता	✓ होता है

- ✓ Note: If a user-defined function, such as hira() is **declared** after the main() function, **an error will occur**. It is because C++ works from **top to bottom**; which means that if the function is not declared above main(), the program is unaware of it:

```

1  #include<iostream>
2  #include<conio.h>
3  using namespace std;
4  int main()
5  {
6      cout << " Hello Hira" << endl;
7      hira();
8      getch();
9  }
10 void hira()
11 {
12     int a,b;
13     cout << " Enter two number = " ;
14     cin >>a>>b;
15     cout << " sum of two number is = " <<a+b;
16 }
17

```

error (Reason :- Not function declaration)

